Adam JINGHAO CHEN

Data Analyst | SQL | Python | Power BI | Tableau
Surrey, BC | Adam_CJH@outlook.com | 604-374-4620 | Portfolio: https://adamchenjinghao.notion.site/

Data Analyst with hands-on experience in data wrangling, analytical modeling, interactive dashboard development, and building and deploying automated data pipelines in cloud environments. Has working knowledge of data engineering, AI, and machine learning. Combines business experience with data analysis and stakeholder communication to translate business questions into clear metrics, visualizations, and practical insights. Contributed to the development of a community-services application for a research project later awarded a $359,928 NSERC grant.

Skills

SQL, Python (Pandas), Excel (Power Query, Power Pivot), Power BI (DAX, Semantic Models), Tableau, Streamlit, Microsoft Fabric, PySpark, PostgreSQL, dbt, Airflow, dlt, MinIO, scikit-learn, NLTK, LangChain, Google Cloud Platform (GCP), Docker, Git, GitHub Actions, Claude Code, Requirements Gathering, Stakeholder Communication, Process Improvement.

Project

Financial Market Analytics & BI Platform
Project Repository: `https://github.com/chenjinghao/ms_fabric_project`

- Developed a financial market analytics platform to analyze the day's three most actively traded stocks, identify potential drivers of sharp price and volume movements, and present market data through Power BI.
- Extracted and transformed raw API data into analysis-ready Delta tables using PySpark and a medallion architecture in Microsoft Fabric, applying data cleaning, validation, and aggregation.
- Designed a Power BI semantic model and DAX measures comparing closing prices and trading volumes with 100-day averages to highlight unusual activity and support analysis of market trends.
- Configured a scheduled Data pipeline to orchestrate data ingestion and transformation in a OneLake lakehouse, with holiday checks, API rate-limit controls, resumable partial runs, and Slack channel failure alerts to support reliable data ingestion and recovery from pipeline failures.
- Built and deployed a GCP version of the platform, running Airflow and dbt in Docker on a virtual machine for daily incremental PostgreSQL updates and hosting a Streamlit dashboard on Cloud Run: `https://www.jinghaodata.engineer/`.

Professional Experience

Research Assistant | Langara College | Vancouver, BC | Jan 2025 – Jul 2025

- Extended a coursework prototype into a functional Streamlit application for shelter and its users, contributing to the Real-Time Access to Essential Services for Communities project later awarded a $359,928 NSERC grant..
- Applied Agile principles to application development, communicating with stakeholders to clarify requirements and iteratively refine features and workflows based on their feedback.
- Built a LangChain-powered AI chatbot that translated natural-language questions into database queries and returned answers in plain language, with safeguards to reject out-of-scope requests.
- Developed role-specific dashboards to visualize shelter reviews processed with natural language processing (NLTK), enable updates to shelter information, and trigger automated workflow solutions to simplify routine administration for shelter staffs.
- Deployed the application to the cloud with CI/CD via GitHub Actions, using GitHub Student Developer Pack benefits to keep the application running online at little to no cost throughout the research period.

Course Assistant | Langara College | Vancouver, BC | Sep 2024 – Jul 2025

- Reviewed assignments across two computer science courses and provided individualized feedback on code logic and analytical methods; coordinated and hosted a capstone project competition with faculty and industry judges.

Licensed Financial Consultant | IPP Financial Advisers Pte. Ltd. | Singapore | Nov 2018 – Nov 2020

- Analyzed client financial information, investment portfolios, and risk profiles to develop actionable investment and insurance recommendations aligned with client goals and market conditions.
- Explained financial products, portfolio risks, and market trends in clear language, helping clients compare alternatives and make informed financial decisions.

Education

Langara College | Post-Degree Diploma in Data Analytics | Vancouver, BC, Canada — Jan 2024 - May 2025
University of Sydney | Bachelor of International and Global Studies | Sydney, NSW, Australia — July 2012 - July 2015

Certifications

- Microsoft Certified: Power BI Data Analyst Associate — Nov 2026
- Microsoft Certified: Fabric Analytics Engineer Associate — Apr 2026
- Claude With Google Vertex AI — Mar 2026
- HackerRank SQL (Advanced) Certificate — Jul 2025

Publication

Efficient Machine Learning Model Deployment in Clinical Decision Support Systems

- **DOI:** 10.1109/CCECE59415.2024.10667100